

Research Interests

Humanoid Robot Learning; Reinforcement Learning and Learning-Based Control; Motion Priors and Imitation Learning; Unified Skill Policies; Whole-Body Control; Contact-Rich Physical Skills; Sim-to-Real Transfer

Education

Nanjing University of Posts and Telecommunications (NJUPT)

B.Eng. Candidate in Artificial Intelligence | Expected Graduation: 2028

GPA: 4.08/5.0 | Ranked 3/133 in Artificial Intelligence

Research Experience

Humanoid Robot Learning and Motion Prior Research

Sep 2026–Present

The Chinese University of Hong Kong | Research Intern (Remote) | Supervised by Dr. Zike Yan (Tsinghua AIR) and Prof. Zhongyu Li (CUHK MAE)

- Participating in research training on humanoid robot learning, whole-body control, motion imitation and motion priors, and task adaptation through literature review, paper presentations, research-question formulation, and experiment planning.
- Reviewed RoboNaldo, Humanoid Goalkeeper, PAiD, PGMPT, SMP, ADD, ConsMimic, and CMDP, analyzing conflicts between task objectives and motion imitation or priors, conditional relaxation, and multi-objective formulations.
- Exploring unified policies, skill-transition and OOD issues from discontinuous motion data, and human-like motion guidance beyond joint-level tracking errors; planning reproduction of SMP and soccer-related baselines.

Reinforcement Learning for Dynamic Humanoid Skills

2026–Present

Independent Project | Booster T1 Locomotion, Kicking, and Conditional AMP Exploration

- Implemented and modified reinforcement-learning environments for a 23-DoF Booster T1 humanoid using MuJoCo, MJLab, and Warp; trained locomotion and kicking skills with PPO.
- Configured 8,192 parallel environments and implemented observation and action spaces, ball-reset logic, contact detection, reward shaping, termination conditions, and curriculum learning.
- Established a training and evaluation workflow for approach, stance adjustment, forward kicking, and recovery; analyzed contact quality, ball velocity, target-direction error, termination causes, and training stability. High-precision kicking accuracy remains under optimization.
- Investigating Conditional AMP and motion-prior approaches for expert-motion and policy-condition alignment, skill generalization, and training stability.

Robot Hand-Eye Calibration with 3D Sensing and Geometric Constraints

2026

Nanjing University of Posts and Telecommunications | Research Project | Manuscript in Preparation

- Conducted hand-eye calibration research for industrial robot and 3D vision systems using an ABB IRB-120/IRC5 and Keyence LJ-X8000A/LJ-X8060 sensors, including robot motion, I/O triggering, profile acquisition, and system integration.
- Designed and evaluated a multi-stage calibration workflow with orthogonal geometric pre-calibration, dynamic point-cloud screening, refinement optimization, and independent testing.
- Collected multi-pose, multi-edge, and multi-point data and helped address practical system issues including robot zeroing, SMB battery maintenance, and trigger synchronization. Published patent application: CN121821412A; manuscript in preparation.

Competition Engineering Experience

Autonomous Aiming Device

2025

National Undergraduate Electronic Design Contest | Vision Perception and Target Tracking | Three-Member Team | Jiangsu Provincial Third Prize

- Built a real-time visual-processing pipeline with a Raspberry Pi and industrial camera; responsible for target detection, target-center localization, and continuous tracking.
- Implemented color-threshold segmentation, contour extraction, and circle detection in OpenCV with approximately 30 Hz feedback; tuned thresholds and target-selection logic for vehicle and gimbal motion, then integrated the pipeline with gimbal and actuator-control modules for closed-loop aiming.

Technical Skills

Programming and Engineering: Python, C++, Git/GitHub, Linux

ML and Robot Learning: PyTorch, PPO, reward design, curriculum learning, humanoid motion control, motion imitation, motion priors, Conditional AMP

Simulation, Robotics, and Vision: MuJoCo, MJLab, Warp, ABB RAPID, robot calibration, 3D sensing, point-cloud processing, OpenCV, target tracking, camera-control integration

Honors

- Third Place, RoboCup China Open 3D Simulation League, 2025.
- Third Place, Jiangsu Provincial Robot Competition, 3D Simulation Group, 2025.